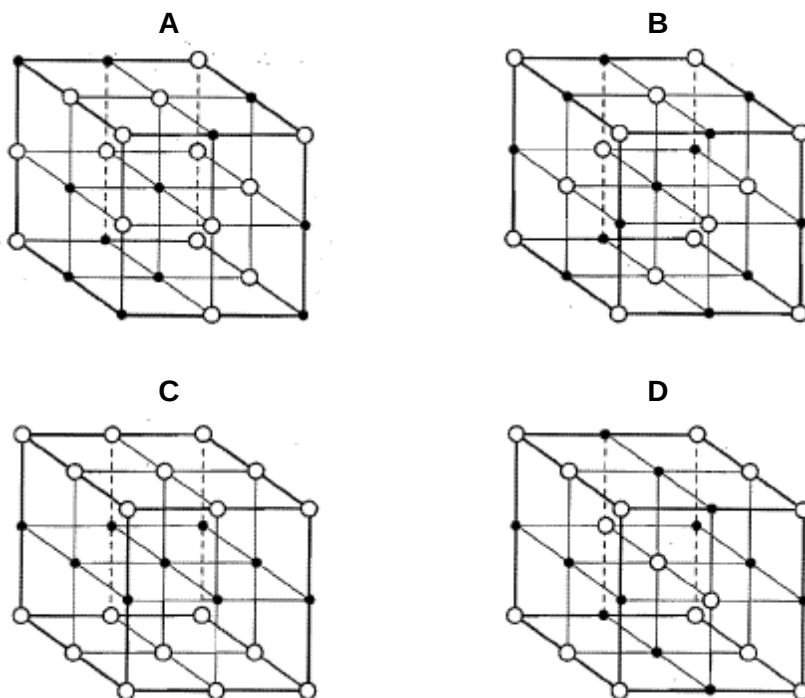


Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and shade your choice on the answer sheet provided.

- 1** Which of the following contains 6.02×10^{23} of the stated particles?
- A** electrons in 1 g of hydrogen gas
B hydrogen ions in 1 dm³ of 1 mol dm⁻³ aqueous ethanoic acid
C chlorine molecules in 35.5 g of chlorine gas
D oxygen atoms in 22.4 dm³ of oxygen gas at s.t.p.
- 2** When 10 cm³ of a hydrocarbon was burnt in excess oxygen, the volume of the mixture decreased by 25 cm³. Upon passing the resultant mixture through a solution of sodium hydroxide, the volume decreased further by 30 cm³. All volumes were measured at room temperature and pressure.
- What is the formula of the hydrocarbon?
- A** C₂H₆ **B** C₃H₆ **C** C₃H₈ **D** C₄H₈
- 3** *Use of the Data Booklet is relevant in this question.*
- Some isotopes are unstable and undergo radioactive decay naturally. In one type of radioactive decay, a neutron decomposes to form a proton, which is retained in the nucleus, and an electron, which is expelled from the nucleus.
- Which change describes a process of this sort?
- A** $^{11}\text{C} \rightarrow ^{12}\text{C}$ **B** $^{111}\text{I} \rightarrow ^{111}\text{Te}$
C $^{76}\text{Br} \rightarrow ^{76}\text{Kr}$ **D** $^{76}\text{Br} \rightarrow ^{75}\text{Br}$
- 4** Element **E** has a high melting point of 1287 °C and is a good conductor of electricity in the solid state. The chloride of **E** has a melting point of 399 °C, does not conduct electricity in both the solid and liquid state, and is soluble in benzene.
- What could element **E** be?
- A** Beryllium **B** Silicon **C** Copper **D** Lithium

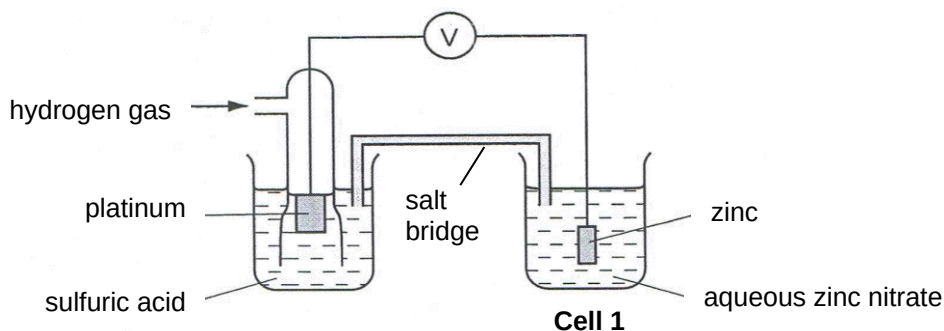
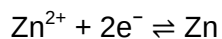
- 5 Which diagram best represents the structure of solid potassium chloride?



- 6 Which statement helps to explain why calcium and chlorine form CaCl_2 rather than CaCl ?
- A** More energy is released in forming chloride ions from chlorine molecules in the formation of $\text{CaCl}_2(\text{s})$ than in the formation of $\text{CaCl}(\text{s})$.
- B** Less energy is required to remove one electron from the calcium atom than to remove two electrons.
- C** More energy is released when $\text{CaCl}(\text{s})$ is formed from its elements than when $\text{CaCl}_2(\text{s})$ is formed from its elements.
- D** The lattice energy of $\text{CaCl}_2(\text{s})$ is more exothermic than that of $\text{CaCl}(\text{s})$.

- 7 *Use of the Data Booklet is relevant in this question.*

The diagram below shows the apparatus used to measure the $E^\ominus$ for the following half-reaction.



Which statement about the above setup is **not** correct?

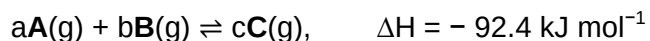
- A Hydrogen gas is supplied at a pressure of 1 atm.
 - B E.m.f. of the cell is measured when the current is effectively zero.
 - C Concentration of sulfuric acid used is 1.0 mol dm^{-3} .
 - D Electrons flow from **Cell 1** to the standard hydrogen electrode.
- 8 *Use of the Data Booklet is relevant to this question.*

When aqueous solutions of copper(II) sulfate and potassium iodide are mixed, a white precipitate is formed in a brown solution. On addition of aqueous sodium thiosulfate to this mixture, the brown colour disappears and the white precipitate remains.

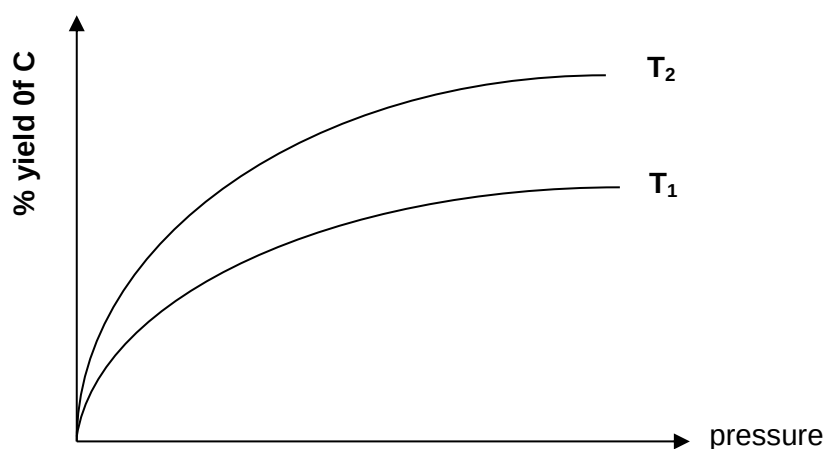
What is the standard cell potential of the reaction of potassium iodide with copper(II) sulfate?

- A +0.39
- B +0.20
- C -0.20
- D -0.39

- 9 Gases **A** and **B** can react to give another gas **C** according to the following equation:



The graph below represents the variation of the equilibrium yield of product **C** with changes in pressure at two different temperatures, T_1 and T_2 .



What can be deduced from the graph regarding the two temperatures, and the magnitude of the stoichiometric coefficients for gases **A**, **B**, and **C**?

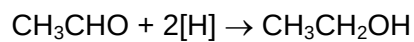
	temperature	stoichiometric coefficients
A	$T_2 < T_1$	$c < a + b$
B	$T_2 < T_1$	$c > a + b$
C	$T_2 > T_1$	$c < a + b$
D	$T_2 > T_1$	$c > a + b$

- 10 The pH of human blood is maintained at around 7.40.

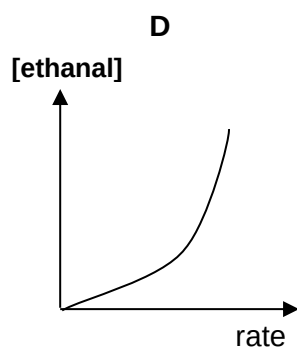
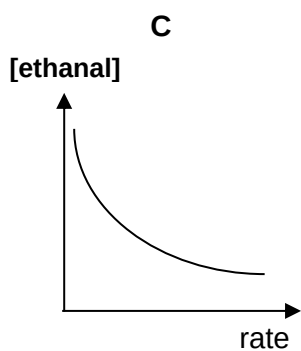
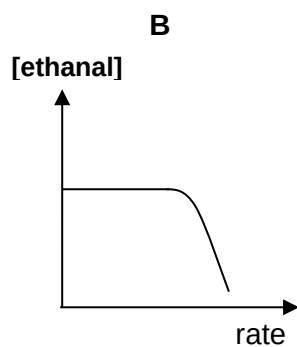
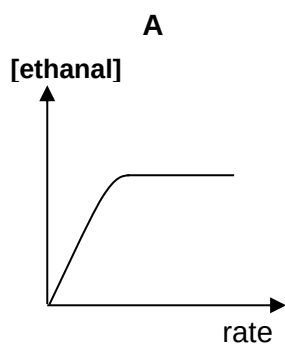
Which ion or molecule present in the human body will remove additional $\text{H}^+(\text{aq})$ ions present in blood to keep the pH constant?

- A** CO_3^{2-} **B** HCO_3^- **C** H_2CO_3 **D** PO_4^{3-}

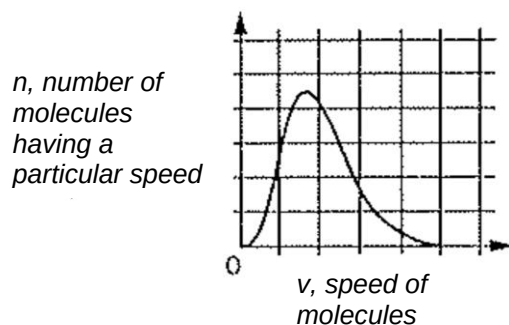
- 11 The enzyme *alcohol dehydrogenase* catalyses an important step in the production of ethanol by fermentation.



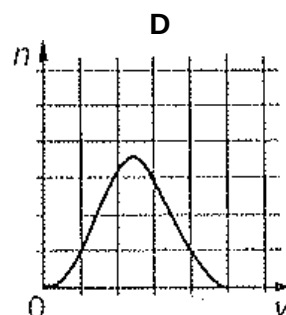
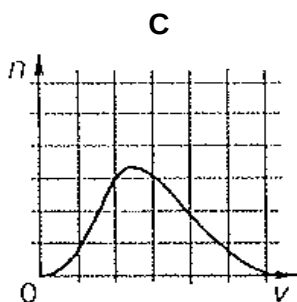
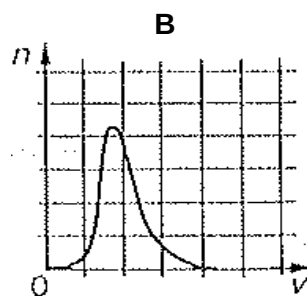
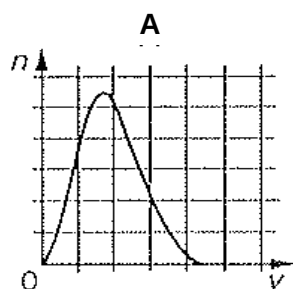
Which graph represents this enzyme-catalysed reaction?



- 12 The diagram below applies to nitrogen at 273K.



Which diagram is likely to apply to nitrogen at 473K?



- 13 An element in Period 3 has the following properties:

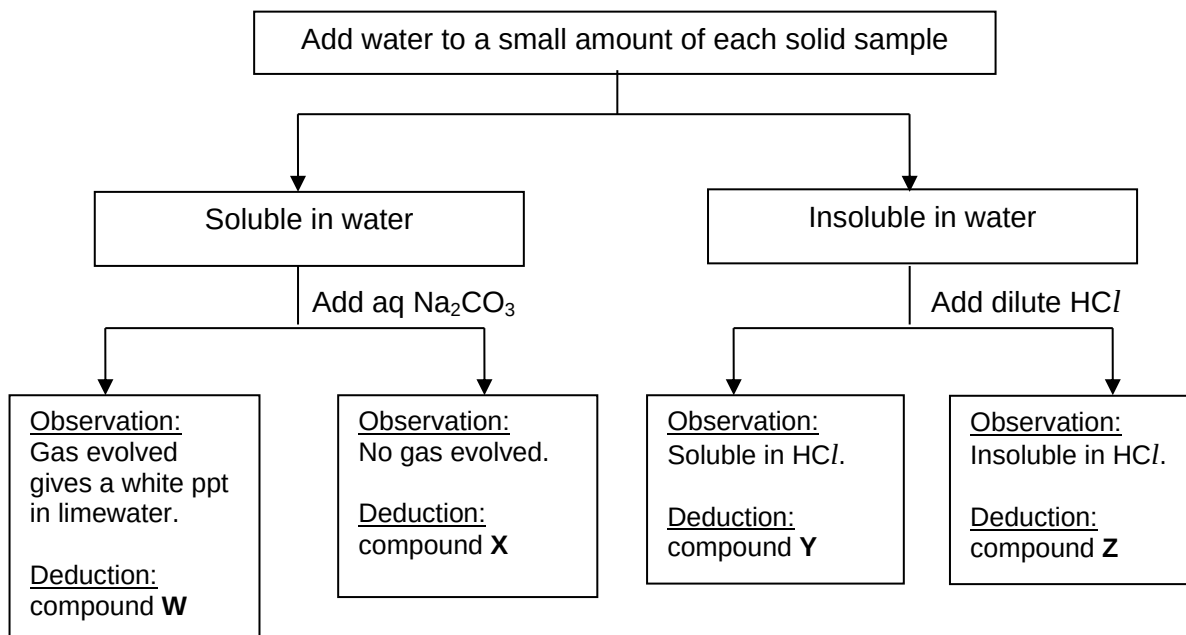
- Its first ionisation energy is larger than both the elements before and after it.
- It conducts electricity.
- It does not react with cold water but can react with steam to give a gas.

Which statement about this element is **not** correct?

- A** It has a high melting and boiling point.
- B** Its oxidation state in compounds usually follows the group number it belongs to in the Periodic Table.
- C** It forms a chloride that produces a solution that is slightly acidic when added to water.
- D** It forms an oxide that produces a solution that is acidic when added to water.

- 14 The labels of four containers containing Na_2O , Al_2O_3 , SiO_2 and P_4O_{10} were detached from the containers.

Using the flowchart given below, identify compounds **W**, **X**, **Y** and **Z**.



	W	X	Y	Z
A	Na_2O	Al_2O_3	SiO_2	P_4O_{10}
B	P_4O_{10}	Na_2O	Al_2O_3	SiO_2
C	SiO_2	P_4O_{10}	Na_2O	Al_2O_3
D	Al_2O_3	SiO_2	P_4O_{10}	Na_2O

- 15 Which statement correctly **defines** a transition element?

- A Transition elements exhibit more than one oxidation state in their compounds.
- B Transition elements form many coloured compounds.
- C Transition elements form one or more stable ions with partially filled d-orbitals.
- D Transition elements or their compounds are widely used as catalysts.

- 16** Three different cobalt(III) complexes with the formula $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ can be prepared by dissolving cobalt(III) chloride in liquid ammonia under various conditions. Different proportions of chloride are precipitated when each of the complexes is treated with silver nitrate.

complex	number of moles of AgCl precipitated per mole of complex	does the complex have a dipole moment?
I	2	yes
II	1	yes
III	1	no

What are the correct structures for complexes **I**, **II** and **III**?

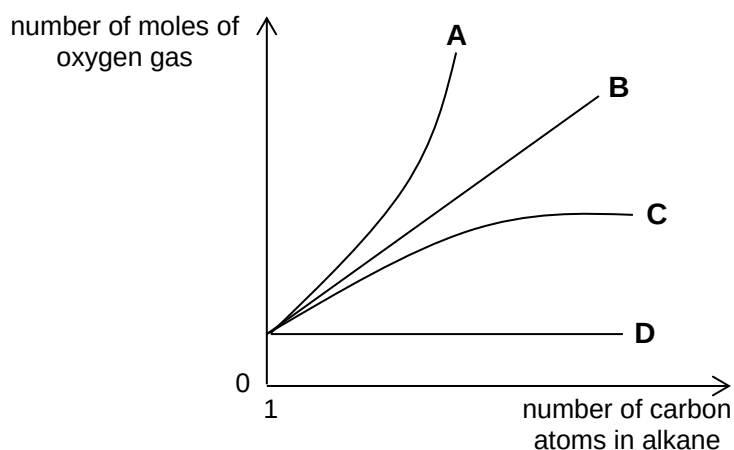
	I	II	III
A			
B			
C			
D			

- 17 What is the total number of isomers for a compound with the molecular formula $C_2H_{6-n}Cl_n$, where n can be any integer from 1 to 6?

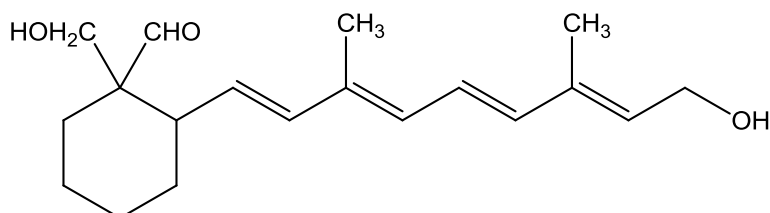
A 6 B 8 C 9 D 10

- 18 The complete combustion of alkanes to produce carbon dioxide and water is an important exothermic reaction.

Which line on the graph shows the relationship between the number of carbon atoms in the alkane and the number of moles of oxygen gas needed for complete combustion of the alkane?



- 19 Vitamin A has multiple functions. It is important for growth and development, for the maintenance of the immune system and good vision. The structure of a Vitamin A derivative recently discovered is shown below.



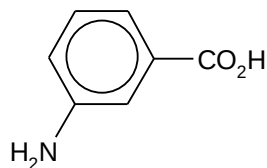
Vitamin A derivative

The derivative is reacted with excess hydrogen gas in the presence of platinum.

What is the total number of stereoisomers in the product?

A 4 B 8 C 16 D 32

- 20 3-aminobenzoic acid is used as a drug against skin disorders.

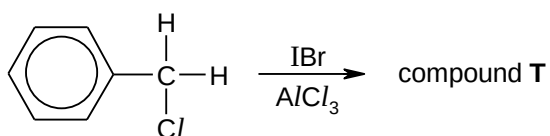


3-aminobenzoic acid

Which reaction sequence is a possible means of its preparation?

- A** $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow{\text{MnO}_4^- / \text{H}^+}$ $\xrightarrow[\text{conc HCl}]{\text{Sn}}$
- B** $\xrightarrow{\text{Cr}_2\text{O}_7^{2-} / \text{H}^+}$ $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow[\text{conc HCl}]{\text{Sn}}$
- C** $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow[\text{conc HCl}]{\text{Sn}}$ $\xrightarrow{\text{H}_2\text{O} / \text{H}^+}$
- D** $\xrightarrow{\text{H}_2\text{O} / \text{H}^+}$ $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow{\text{NaBH}_4}$

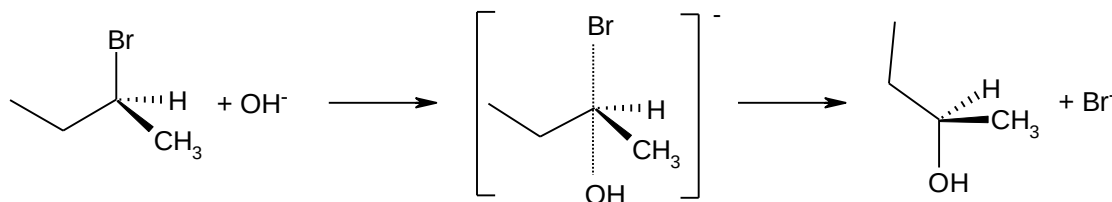
- 21 Bromine, in the presence of anhydrous aluminium chloride, undergoes a substitution reaction with benzene forming bromobenzene. When chloromethylbenzene reacts with IBr, in the presence of anhydrous aluminium chloride, a similar reaction occurs and compound **T** is formed.



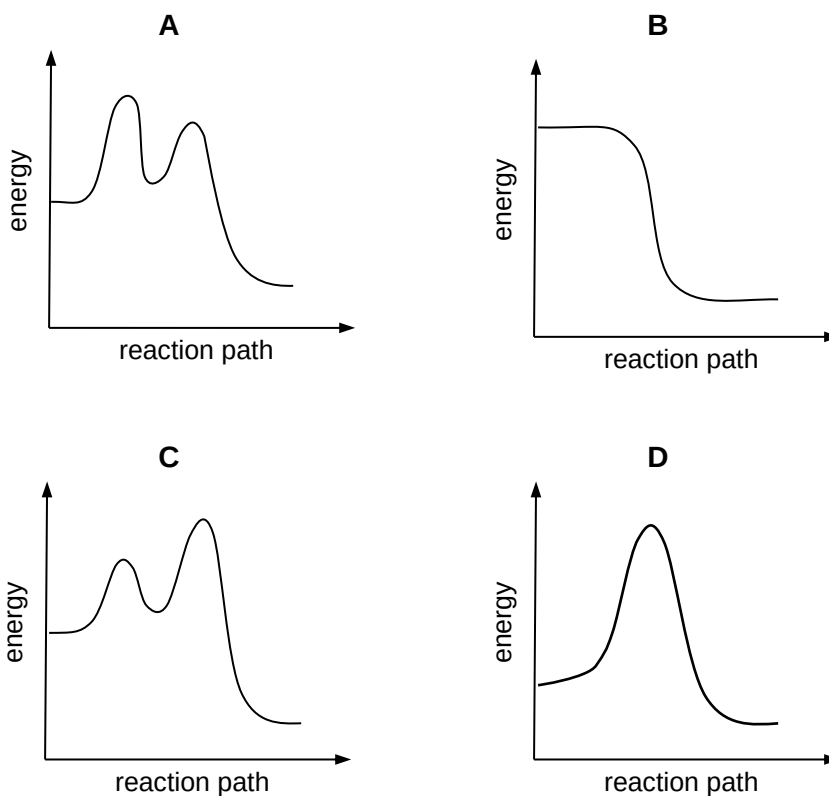
Which product will be formed when compound **T** is heated under reflux with aqueous sodium hydroxide?

- A**
- B**
- C**
- D**

- 22 The mechanism for the hydrolysis of 2-bromobutane is shown.



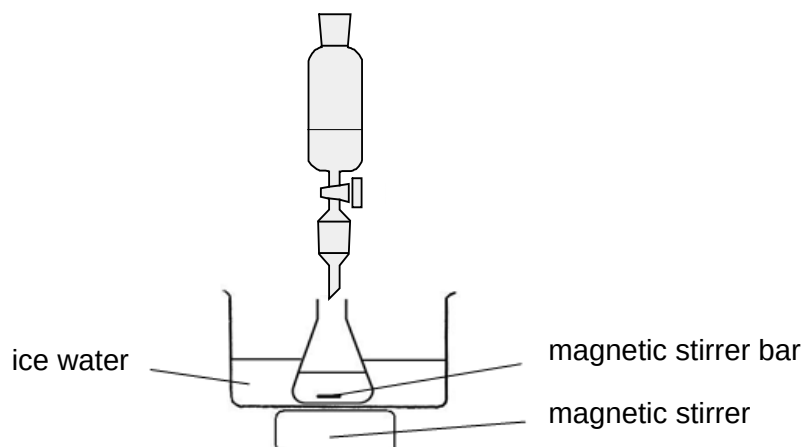
Which diagram represents the reaction profile for this mechanism?



- 23 Dichlorodifluoromethane, CCl_2F_2 , is widely used in aerosol propellants and as a refrigerant. Which statement helps to explain why dichlorodifluoromethane is chemically inert?

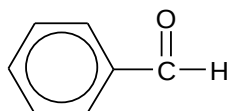
- A The carbon-fluorine bond energy is large.
- B The carbon-fluorine bond has a low polarity.
- C Fluorine compounds are non-flammable.
- D Fluorine is highly electronegative.

- 24 The diagram shows an experimental setup used in a laboratory.

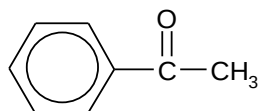


Which preparation could this setup be used for?

- A 2-bromophenol from phenol and bromine in tetrachloromethane
 - B Ethanol from ethene and concentrated sulfuric acid
 - C Propan-2-ol from 2-chloropropane and aqueous sodium hydroxide
 - D Ethyl methanoate from ethanol, methanoic acid and concentrated sulfuric acid
- 25 Which reagent may be used to distinguish between benzaldehyde and 1-phenylethanone?



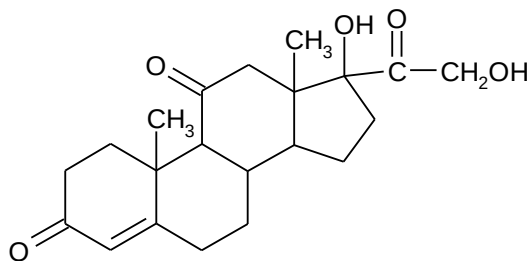
benzaldehyde



1-phenylethanone

- A Brady's reagent
- B Fehling's reagent
- C Na metal
- D I_2 in NaOH(aq)

- 26 Cortisone is an anti-inflammatory hormone.

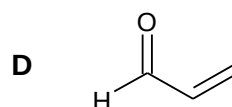
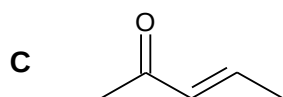
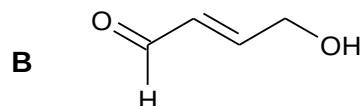
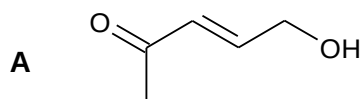


Cortisone

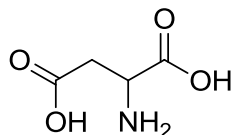
Which statement about cortisone is true?

- A** When treated with cold dilute KMnO_4 , it forms a compound containing two hydroxyl groups.
- B** When warmed with aqueous alkaline iodine, a yellow precipitate is observed.
- C** When treated with NaBH_4 in the presence of methanol, it forms a compound containing five hydroxyl groups.
- D** When treated with an excess of hot concentrated acidified KMnO_4 , it forms a compound containing seven carbonyl groups.
- 27 Compound **S** produces an orange precipitate when added to 2,4-dinitrophenylhydrazine but no precipitate with Tollens' reagent. When heated with dilute acidified potassium manganate(VII), effervescence of a gas is observed.

What is the structural formula of compound **S**?



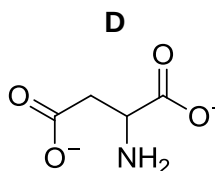
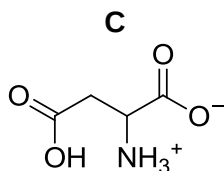
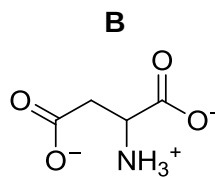
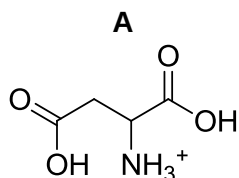
- 28 Aspartic acid is found in vegetable sources such as sprouting seeds.



aspartic acid

Aspartic acid has three pK_a values of 2.10, 3.86, 9.82.

What is the structure of the major species present in a solution of aspartic acid at pH 3?

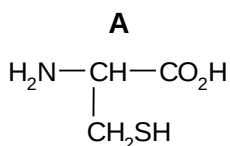


- 29 Trypsin is an enzyme that converts proteins into peptides and amino acids.

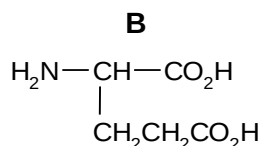
What type of reaction does trypsin catalyse?

- | | |
|-------------------------------------|----------------------------|
| A hydrolysis | B elimination |
| C electrophilic substitution | D homolytic fission |

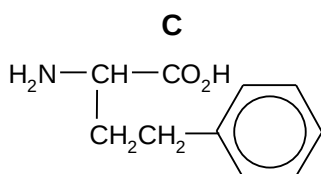
- 30 Which amino acid residue cannot form a hydrogen bond to stabilise the peptide α -helix and so causes a "bend" in the regular shape?



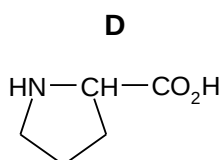
cysteine



glutamic acid



phenylalanine



proline

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to pick a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31** 10 dm³ of gas **S** weighs 0.5 g and 10 dm³ of gas **T** weighs 2.5 g under the same conditions of temperature and pressure.

Assuming ideal gas behaviour, which statements are correct?

- 1** The average kinetic energy of the molecules in gas **S** and gas **T** are the same at the same temperature.
- 2** The ratio of the M_r of **S** to **T** is 1 : 5.
- 3** The number of molecules of **T** in 10 dm³ is five times the number of molecules of **S** in 10 dm³.

- 32** Solutes dissolve in solvent to form solutions. If a semi-permeable membrane, that allows solvent molecules to pass through it, is placed between a pure solvent and a solution containing a solute in that solvent, pure solvent only will go through the membrane into the solution.

In which processes will ΔS be positive?

- 1** Dissolving the solute in a solvent
- 2** The evaporation of the solvent from the solution
- 3** The passage of the solvent through a semi-permeable membrane during osmosis

- 33** *Use of the Data Booklet is relevant to this question.*

When 290 C of electricity is passed through a molten compound of a metal, 1.00×10^{-3} mol of atoms of the metal is deposited at the cathode.

What could the metal be?

- 1** Silver
- 2** Iron
- 3** Chromium

- 34** Three substances, **L**, **M** and **N**, react to give a coloured substance **P**. The concentration of **P** produced can be monitored using a colourimeter. The time taken, t , for various initial concentrations of **L**, **M** and **N** to produce a $1.0 \times 10^{-3} \text{ mol dm}^{-3}$ of **P** are given in the following table:

experiment	[L] / mol dm^{-3}	[M] / mol dm^{-3}	[N] / mol dm^{-3}	t / s
1	0.5	0.05	0.05	176
2	2.0	0.05	0.05	176
3	1.0	0.10	0.05	44
4	1.0	0.10	0.10	22

Which statements about the reaction are correct?

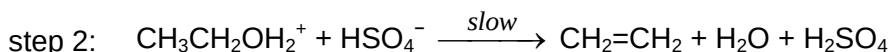
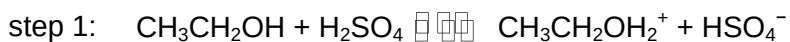
- 1 The rate equation for the reaction can be written as $\text{rate} = k [\text{M}]^2 [\text{N}]$.
 - 2 The unit of the rate constant is $\text{mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$.
 - 3 The half-life of the reaction with respect to **N** is independent of the concentration of **L** and **M**.
- 35** Which properties of the Group II elements or their cations show a decreasing trend down the group?
- 1 Magnitude of hydration enthalpy of cations
 - 2 Reducing power of elements
 - 3 Atomic radius of elements
- 36** Concentrated sodium chloride (brine) is electrolysed by using inert electrodes in a cell which is stirred so that the products of electrolysis are able to react. The cell is kept cold. Which substances are among the final products?
- 1 Hydrogen gas
 - 2 Sodium chlorate(I)
 - 3 Sodium chlorate(V)
- 37** Which reactions are examples of nucleophilic substitution?
- 1 $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH}(\text{aq})$
 - 2 $\text{CH}_3\text{CN} + \text{LiAlH}_4$
 - 3 $\text{C}_6\text{H}_5\text{OH} + \text{Br}_2(\text{aq})$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

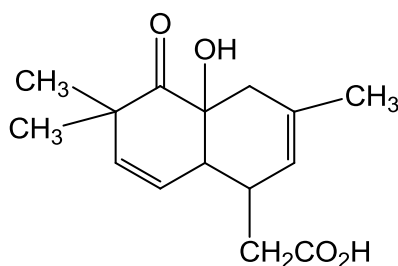
No other combination of statements is used as a correct response.

- 38** The dehydration of ethanol to form ethene is thought to involve the following steps.



Which statements about the reaction are correct?

- 1** The overall order of reaction is two.
 - 2** H_2SO_4 is a catalyst in this reaction.
 - 3** 2-methylpropan-2-ol is more likely to proceed via this mechanism than ethanol as the formation of the carbocation is more favoured.
- 39** Which reactions will produce a gas?
- 1** $\text{C}_6\text{H}_5\text{OH} + \text{PCl}_5(\text{s})$
 - 2** $\text{CH}_3\text{CH}_2\text{OH} + \text{Na}(\text{s})$
 - 3** $\text{CH}_3\text{CO}_2\text{H} + \text{Na}_2\text{CO}_3(\text{aq})$
- 40** Which statements about compound **U** shown below are correct?



Compound **U**

- 1** 1 mol of **U** requires 1 mol of aqueous NaOH for complete reaction.
- 2** **U** reacts with excess LiAlH_4 to form a compound with 16 stereoisomers.
- 3** **U** reacts with hot acidified KMnO_4 to give a product with four carboxylic acid functional groups.

~ END OF PAPER ~